3 Fig. 3 shows a triangle PQR. The vector $\overrightarrow{PQ}$ is $\mathbf{i}+7\mathbf{j}$ and the vector $\overrightarrow{QR}$ is $4\mathbf{i}-12\mathbf{j}$.

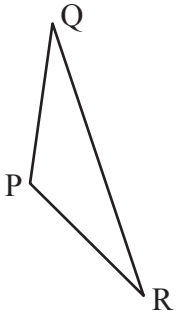


Fig. 3

(a) Show that the triangle PQR is isosceles. **[3]**

The point P has position vector $-3\mathbf{i}-\mathbf{j}$. The point S is added so that PQRS is a parallelogram.

(b) Find the position vector of S. **[2]**